

# Aryan Mathur

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[GitHub](#) | [LinkedIn](#)

## EDUCATION

### Prestige Public School (2021)

Class X Percentage (91.16%)

Indore, India

### Prestige Public School (2023)

Class XII Percentage (79.8%)

Indore, India

### Vellore Institute of Technology (2023-2027)

Computer Science and Engineering (AIML); GPA: 8.31

Bhopal, India

## SKILLS SUMMARY

- **Languages:** Python
- **Libraries:** NumPy, Pandas, Xgboost, Matplotlib, Seaborn
- **AI/ML/DL & Framework:** TensorFlow, PyTorch, Keras, Scikit-Learn
- **Developer Tools:** Visual Studio Code, Google Collab, Jupyter, Git, GitHub

## RELEVANT EXPERIENCE AND PROJECTS

### Sign Language Detection Model using Computer Vision:

March 2025

- Developed a real-time Sign Language Detection System using computer vision and deep learning techniques to recognize hand gestures from image/video input, enabling effective human-computer interaction and accessibility support.
- Implemented data processing, feature extraction, and model training using CNN-based architectures, achieving accurate gesture classification and improving recognition performance through model optimization and evaluation.

### Crop Prediction and Price Recommendation System:

August 2025

- Developed a machine learning-based crop recommendation and price prediction system using historical agricultural and environmental data to assist farmers in selecting optimal crops and forecasting market trends.

### Temporal and Contradiction-Aware Retrieval-Augmented Generation:

Ongoing

- A Retrieval-Augmented Generation (RAG) system designed to address two failure modes standard RAG pipelines ignore: outdated information and unresolved contradictions between retrieved sources. The system incorporates temporal-aware retrieval, which ranks and filters documents by relevance to a query's implied time scope, and contradiction detection using natural language inference models to identify conflicting claims across sources.
- Rather than silently blending conflicting information into a single answer, the system applies an explicit resolution policy, preferring recent sources, presenting both sides, or abstaining, while flagging detected conflicts to the user.
- Evaluation combines automatic benchmarking against recent comparable systems on datasets including TimeQA, FEVER, and WikiContradict, alongside a human study measuring whether transparency about conflicts improves user trust and comprehension.

## CERTIFICATES

### Python Essentials, VITYARTHI

May 2023

- Certified in Python programming, data structures.

### Fundamentals of AI and ML, VITYARTHI

March 2024

- Learned core AI/ML concepts, model building, and real-world applications.
- Acquired expertise in procedural programming paradigms and associated logical concepts, enhancing capabilities.

### Applied Machine Learning in Python | [LINK](#)

December 2025

- Applied machine learning in Python, focusing on model building and performance evaluation

### Computer Vision, VITYARTHI

February 2026

- Certified in teaching Image Processing, Object Detection, and Deep Learning Fundamentals.
- Emphasizes hands-on projects to build real-world AI vision applications.

## INTERNSHIP

### AIML Internship at Empower Solutions (Indore)

May 2026

- Developed a Flight Fare Prediction model using [Random Forest / XGBoost / LSTMs] and optimized its accuracy through hyperparameter tuning (evaluated via RMSE and MAE).
- Engineered a data preprocessing pipeline utilizing One-Hot Encoding to handle categorical features like airlines, cities, and layover points.
- Built a personalized destination recommendation engine using clustering algorithms to segment users based on search and travel history.